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Further paleomagnetic results from the Silurian of the Yangtze Block and their implications

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Abstract

Silurian redbeds were sampled at two localities in northeastern Guizhou and Yunnan provinces of China for paleomagnetic study. Progressive thermal demagnetization revealed multiple magnetic components from these samples: a low unblocking temperature component (A) of recent origin, a post-folding middle to high unblocking temperature component (B) probably acquired during late Mesozoic time, and a pre-folding high unblocking temperature component (C). Two statistically distinguishable groups of directions were further recognized from the C component with one (C1) having more easterly declination than the other (C2) and both having a of shallow inclination. The paleomagnetic pole from the C1 magnetization coincides with a pre-folding pole reported from the coeval rocks in neighboring southern Sichuan Province; its relation to the apparent polar wander path (APWP) for the Yangtze Block (YB) suggests that the C1 magnetization was probably acquired during the Silurian. By the same argument, the C2 magnetization was probably acquired sometime between the Silurian and Carboniferous. These results reconfirm the equatorial position for the YB during mid-Paleozoic time and clockwise rotation of the YB since at least the mid-Cambrian. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Silurian; Yangtze Platform

1. Introduction

Our understanding of the tectonic evolution of China since the Permian has been greatly improved thanks to intensive geological and paleomagnetic investigations in the last two decades. However, the picture for pre-Permian time is less clear. Paleomagnetic experience indicates that the older the rocks, the more difficult they are to work with. This is because older rocks experience longer and usually more complicated geological history, and their original magnetization is thus often modified by later geomagnetic events and sometimes completely obliterated. For this reason, it should not come as a surprise that paleomagnetic results reported from the Paleozoic or older rocks from each of the major tectonic blocks in China are much fewer than those from younger rocks. A case in point is the Silurian for the Yangtze Block (YB) in South China. The only Silurian paleomagnetic pole for the YB that meets

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modern reliability criteria is the one we reported 10 years ago [1]. This pole indicates an equatorial position for the YB, but resembles the poles derived from Cambrian [2] and Devonian rocks [3], thus causing suspicion of remagnetization of these rocks [3,4]. Therefore, reliable paleomagnetic data are needed to resolve such controversies so that meaningful geological interpretations can be made. In this paper, we report new Silurian paleomagnetic results from the YB and discuss their geological implications.

2. Geology and sampling

The Silurian rocks are widespread on the YB, especially in the southwestern region where type sections for the Silurian System of China have been established. At present, three series and six stages have been proposed for the Chinese Silurian System [5–7], in ascending order: the Longmaxian and Shiniulanian Stages of the Lower Series (Llandoverian), the Baisha'an and Xiushanian Stages of the Middle Series (Wenlockian), and the Guandian and Miaogao'an Stages of the Upper Series (Ludlovian–Pridolian) [7]. In 1987, we studied the paleomagnetism of the Silurian rocks collected from three type sections, and the samples from Rongxi, Xiushan, southern Sichuan Province yielded consistent results [1]. In this study, two more localities were sampled: one in Shiqian County in northeastern Guizhou Province (27.5°N, 108.0°E) and another in Daguan County in northeastern Yunnan Province (27.9°N, 103.9°E) (Fig. 1).

The Silurian sequence near Shiqian is represented by the Leijiatun section about 5 km north of downtown Shiqian. The lithology and fossil record of this section are correlative with the Rongxi section in southern Sichuan [8]. The Lower Silurian Series includes, in ascending order, the Longmaxi, Xiangshuyuan, Leijiatun and Majiaochong formations. The Longmaxi Fm at this section is only 6 m thick, consisting of gray shales, disconformably overlying limestones of the Upper Ordovician Guanyinqiao Fm, and yielding abundant graptolites. The Xiangshuyuan Fm is 68 m in thickness, the lower part of which is composed of yellowish green shales interbedded with limestones and bioclastic limestones, and the upper part of which is composed of gray thick-bedded bioclastic limestones. The Leijiatun Fm is 87 m thick, and consists of yellowish green shales with subordinate bioclastic and nodular limestones. Both the Xiangshuyuan and Leijiatun formations are rich in coral, conodont and brachiopod fossils. The Majiaochong Fm is 60 m in thickness and is composed primarily of yellowish green shales. The Middle Silurian Series is divisible into the Rongxi and Xiushan formations. The Rongxi Fm was previously called the Baisha Fm, and is 208 m in thickness and characterized by purplish red and yellowish green mudstones and siltstones. The overlying Xiushan Fm is 286 m thick, made up of yellowish gray mudstones and muddy siltstones with bioclastic limestone interbeds, and yielding abundant trilobites, brachiopods, conodonts, nautiloids and graptolites. The Upper Silurian is represented by the Huixingshao Fm, which is 65 m thick and is characterized by purplish red silty mudstones and sandstones and disconformably underlies the lower Permian limestones (Qixia Fm). Therefore, as in most parts of the YB, the stratigraphic record of the Upper Silurian at the Leijiatun section is incomplete, reflecting a post-Silurian regional uplift of the YB. Paleomagnetic sampling in the redbeds of the Silurian sequence was carried out along the highway from Shiqian to Kaiyang on three limbs of two adjacent NNE-trending folds (Fig. 1a). At the eastern limb of the syncline, where the Leijiatun section is established, the rock beds dip westward at 44–54°. Five sites (1–5), usually six samples per site, were drilled in the Rongxi Fm and two sites (6 and 7) were in the Huixingshao Fm using a gasoline-powered rock drill and oriented with a magnetic compass. At the western limb of the syncline, seven more sites (8–14) were collected from the Rongxi Fm exposed near a township called Basha. The bedding there dips east–southeastward at 39°. Further west near a township called Benzhuang, five sites (15–19) were drilled in the Rongxi Fm at the western limb of the anticline which dips west–northwestward at 51–54°.

The Silurian sequence near Daguan, northeastern Yunnan, is represented by the Huanggexi sec-

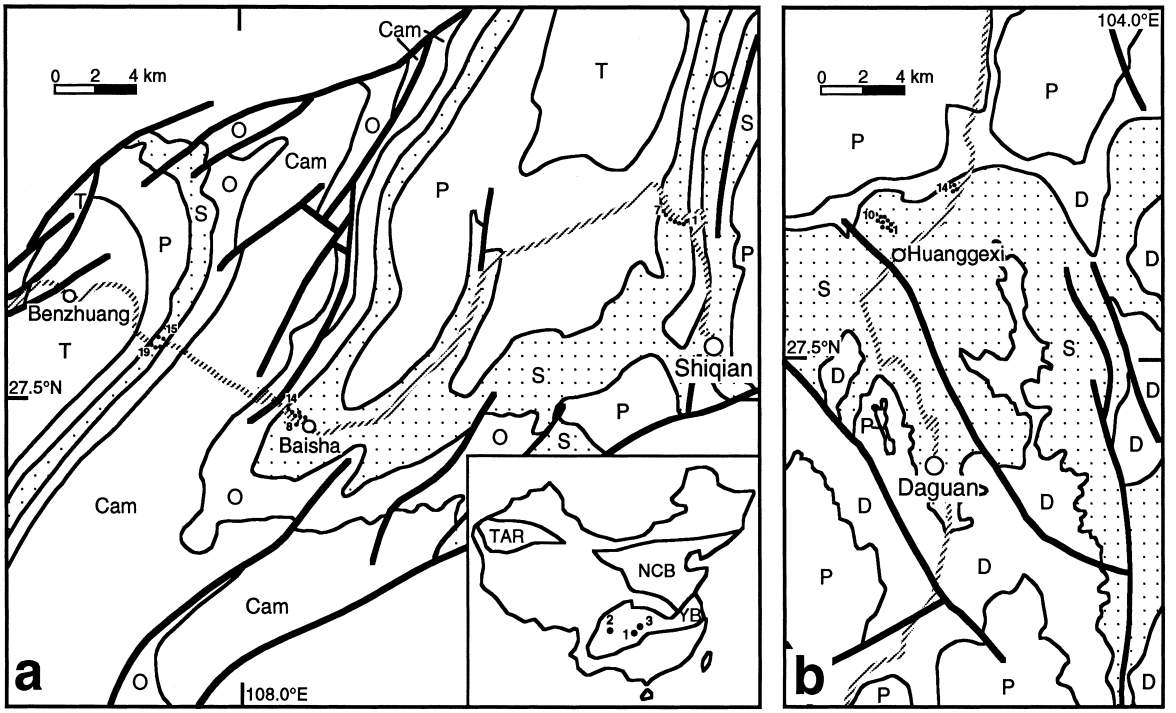


Fig. 1. Geologic sketch maps for Shiqian (a) and Daguan (b). Numbered dots denote sampling sites. Thick lines are faults. Hatched lines are major highways. Strata: Cam = Cambrian, O = Ordovician, S = Silurian (stippled), D = Devonian, P = Permian, T = Triassic. The inset shows three major cratons in China (NCB = North China Block, TAR = Tarim Block, YB = Yangtze Block) and the sampled localities (dots) for the Silurian (1. Shiqian, 2. Daguan, 3. Xiushan [1]).

tion about 10 km north of downtown Daguan (Fig. 1b), which has, in fact, been proposed as a supplementary type section for the Middle Silurian Series of China [5,7]. The section is situated at the northern flank of an open anticline with the Silurian constituting the core and Devonian and Permian rocks cropping out at the flanks. The sequence at this section correlates well with that at the Leijiatun section in Shiqian and is richly fossiliferous, but it is thicker (959 m) and contains more limestones [7,9]. It conformably overlies the Ordovician and disconformably underlies the Devonian rocks. The equivalents of the Rongxi and Huixingshao formations are called the Niugundang and Caidiwan formations, respectively, at this section, and they are also characterized by redbeds. Only the Niugundang Fm, which is 135 m thick and composed chiefly of purplish red calcareous shales with limestone interbeds, was sampled. Ten sites (1–10) were drilled in the calcareous silty shales exposed about 200 m above the highway from Daguan to Yibing, where the rock bedding dips east–northeastward at 13°. Four more sites (11–14) were drilled in a small gully about 4 km further north along the highway, where the bedding attitude changes to dipping northward at 42–43° due to flexure along the strike.

Folding of the Silurian rocks in northeastern Guizhou probably took place sometime between late Jurassic and late Cretaceous. This is inferred from the unconformity between the Upper Jurassic and Upper Cretaceous observed from adjacent southern Sichuan Province [8]. The age of folding in northeastern Yunnan is uncertain, but it should not be earlier than late Cretaceous [9].

3. Results

All the samples were subjected to progressive thermal demagnetization using an ASC or Shon-

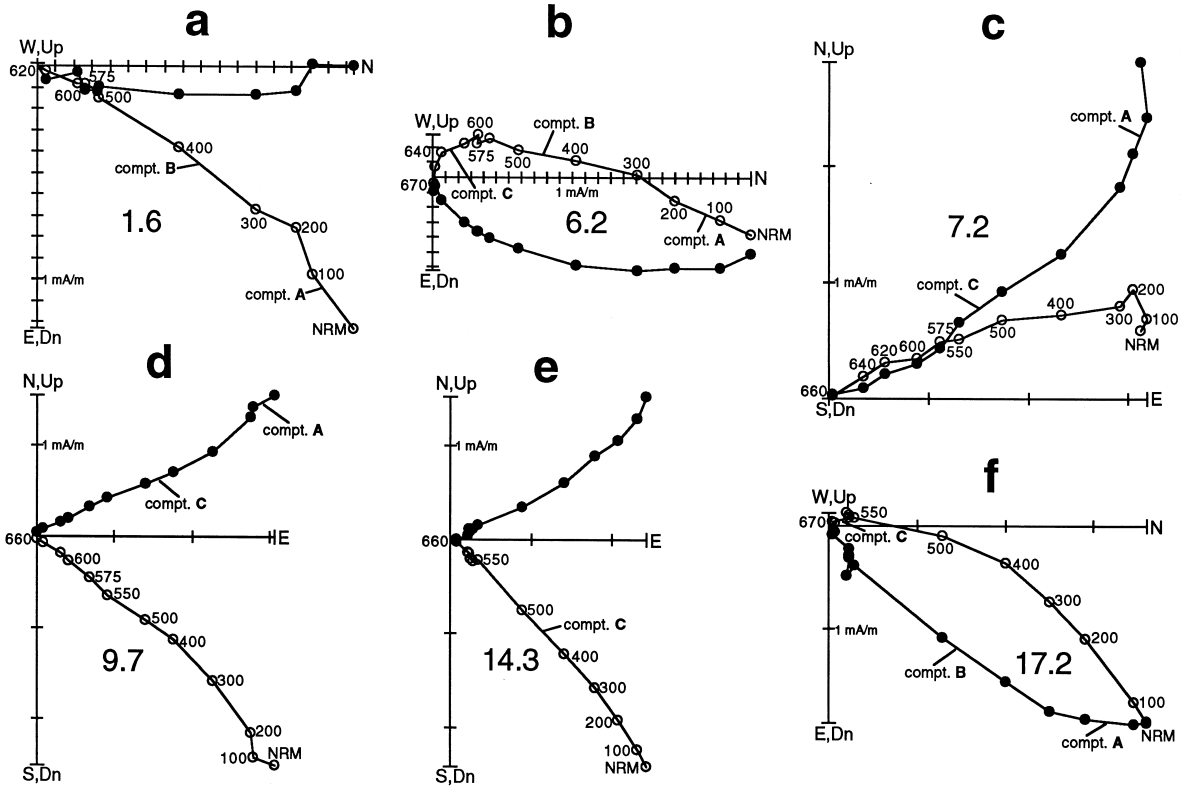


Fig. 2. Representative orthogonal plots of thermal demagnetization (in geographic coordinates) for the samples from Shiqian. Solid (open) circles are projections of vector end points on horizontal (vertical) planes. Treatment levels are in °C. Magnetic components (compt.) are indicated.

stedt thermal demagnetizer. Magnetization was measured in a 2G cryogenic magnetometer. This equipment is housed in a magnetically shielded room at the Paleomagnetic Laboratory of the University of Florida. Demagnetization data were analyzed utilizing standard paleomagnetic analytical techniques [10–13].

3.1. Shiqian, northeastern Guizhou

The natural remanent magnetization (NRM) of the samples from sites 1–5 is weak with initial intensities being in the range from 1.2 to 2.9 mA/m. Samples from the remaining sites are more strongly magnetized with a majority of the initial NRM intensities ranging from 2.5 to 7.0 mA/m. Three magnetic components were recognized. A low temperature component, designated as component A hereafter, was seen in nearly every sample (Fig. 2), and was eliminated at about 200°C. The mean in situ direction of component A ($D/I = 7.5°/45.3°$) is very close to the dipole field direction ($D/I = 0°/46.2°$) at Shiqian; it fails the fold test ($\kappa_s/\kappa_g = 0.20$, see Fig. 3 upper). Thus, component A is apparently a recent overprint. A second component, designated as component B, was observed in the samples from sites 1–6 and 15–19 (Fig. 2a,b and f). It mostly unblocked in the temperature range from 300° to 550°C (Fig. 2b,f), and in some cases persisted over 600°C (Fig. 2a). Component B is northerly or north–northeasterly with shallower inclinations than the present dipole field direction in geographic coordinates, and it also fails the fold test ($\kappa_s/\kappa_g = 0.58$, Fig. 3 middle). A third component, designated as component C, was resolved from most of the samples from all but sites 1–5. Component C dominated the NRM of the samples from sites 7–14:

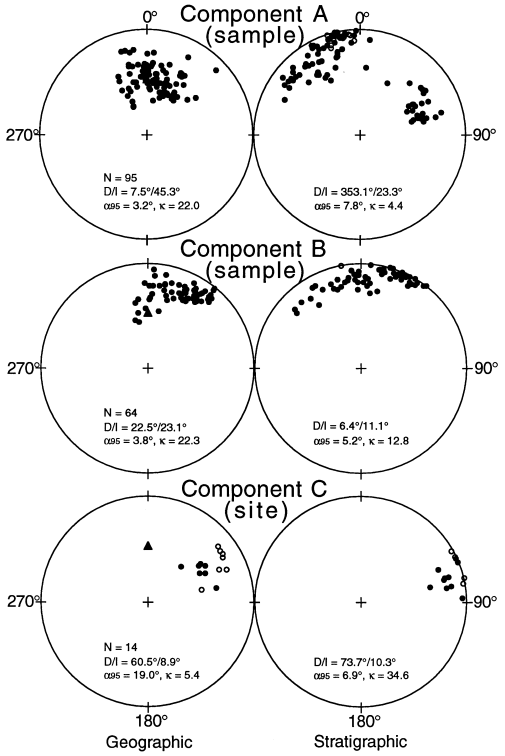


Fig. 3. Equal-area projections of magnetic component directions observed in the samples from Shiqian. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Black triangle denotes the present dipole field direction at Shiqian.

after removal of component A at about 200°C, component C linearly decayed toward the origin of the coordinates up to 660°C (Fig. 2c–e). In the case of sites 6 and 15–19, component C was not resolved until above 600°C. In both cases, the magnetic carrier is apparently hematite, and this is further confirmed by isothermal remanent magnetization (IRM) acquisition and thermal demagnetization of composite IRM experiments (Fig. 4). Because of weak magnetic intensity above 600°C and partially overlapped blocking temperature spectra between components C and B as evidenced by the curved demagnetization trajectory (Fig. 2b,f), the direction of component C in sites 6 and 15–19 could not be adequately determined by using principal component analysis technique [12], so the sector-constrained great circle intersection method [13] was employed (Fig. 2b,f). Although the declinations of the C component resolved from these sites are all east–northeasterly before bedding correction, the inclinations observed from sites 6 to 7 and 15–19 taken from the west-dipping limbs are upward whereas those from the east-dipping limb (sites 8–14) are downward. After correcting for bedding tilt, they all become east–northeasterly with shallow inclinations (Fig. 3 lower), and precision parameter, κ, increases by a factor of 6.4, which is much greater than McElhinny's [14] threshold value (2.55) for a positive fold test at the 99% confidence level. We will further examine this in Section 4. Statistics for component C are given in Table 1.

3.2. Daguan, northeastern Yunnan

The initial NRM intensities of the samples collected from Daguan vary from 1.0 to 8.9 mA/m. Like the samples from Shiqian, three magnetic components were observed. Component A is present in some of the samples and was removed below 200°C (Fig. 5c). Although rather scattered, the A component is largely directed in the Earth's present magnetic field with a mean of $D/I = 12.9°/47.2°$ ($n = 24$ samples, $\alpha_{95} = 11.8°$) in geographic coordinates. Component B dominated the NRM of sites 1–7 and persisted up to 640°C (Fig. 5a,b), and is probably carried by hematite. It is also northerly or northeasterly with shallower inclinations than present dipole direction (mean in situ direction $D/I = 13.0°/36.0°$, $n = 40$ samples, $\alpha_{95} = 3.9°$, see Fig. 6 upper). Because of the uniform bedding attitude, however, a tilt test is not applicable. Component C was dominant and well isolated from sites 8–14 (Fig. 5c,d). Some samples became rather viscous at high demagnetization temperatures due to the very fine grained nature of the rocks (Fig. 5d). In sites 5 and 6, component C was not resolved until above 640°C (Fig. 5b), and could only be determined using the sector-constrained great circle intersection method [13]. Tilt correction only slightly improves the grouping of site mean directions of component C ($\kappa_s/\kappa_g = 1.3$) (Fig. 6 lower). Again, we will further examine this in Section 4. Statistics for component C are listed in Table 2.

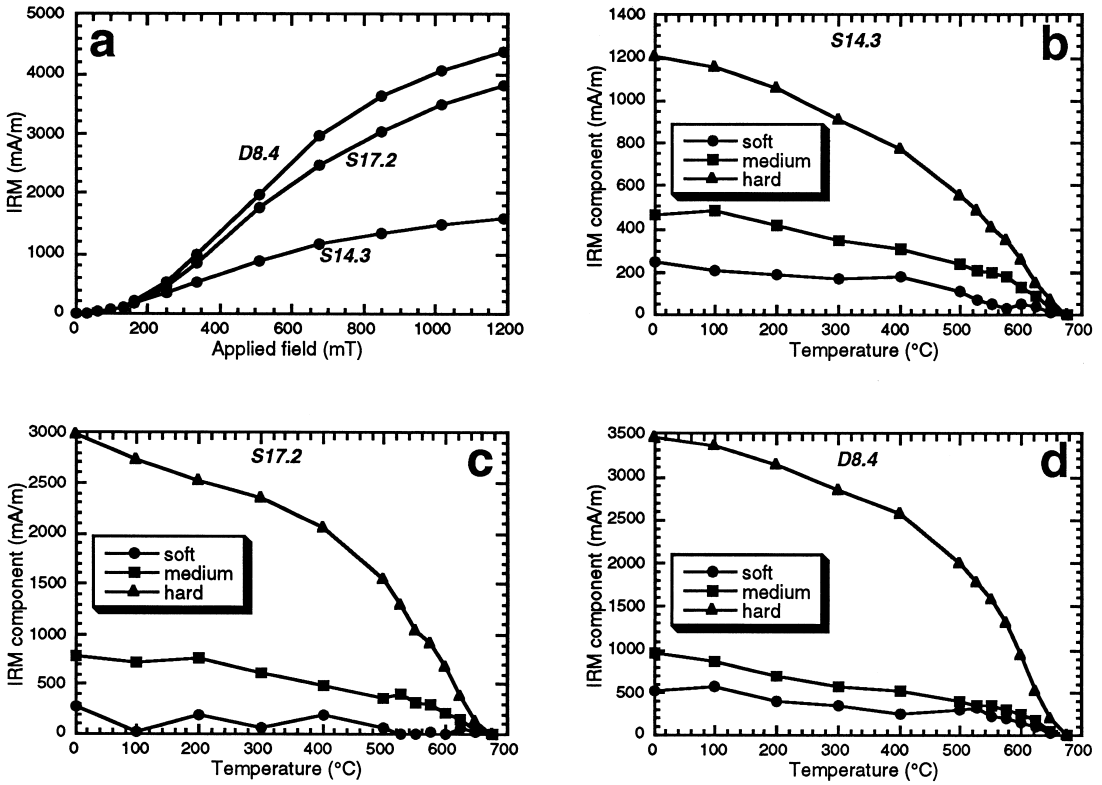


Fig. 4. Progressive IRM acquisition (a) and thermal demagnetization of composite IRM (b–d) for representative samples. Samples S14.3 and S17.2 are from Shiqian, and D8.4 from Daguan. In all the samples studied, IRM intensity gradually increased with increasing DC field but did not reach saturation up to 1.2 T. The composite IRM was produced by sequentially applying 1.2, 0.3 and 0.1 T DC field to z, y and x axes of the sample (following [27]). There is little indication of other magnetic minerals than hematite in the decay of the hard, medium and soft coercivity fractions.

4. Discussion

4.1. Age of magnetizations

As demonstrated above, component A is a recent overprint of the Earth's magnetic field, which is evidenced by its similarity to the dipole field direction at the sampling sites and negative fold test. The fact that the paleomagnetic poles computed from the in situ directions of component A fall close to the present geographic North pole or the Tertiary pole for the YB (Fig. 8) further supports this conclusion. A negative fold test at Shiqian indicates that component B is also a post-folding magnetization. The pole positions derived from the in situ mean directions of the B component from Shiqian and Daguan are indistinguishable from the late Jurassic or late Cretaceous pole for the YB within paleomagnetic uncertainties (Fig. 8), suggesting that component B was probably acquired during late Mesozoic time. The presence of components A and B in these Silurian rocks reinforces the observation that recent to late Mesozoic remagnetization is common in pre-Jurassic rocks on the YB [15–17]. Tilt correction clearly improves the grouping of the site mean directions of the C component at Shiqian (see Fig. 3 lower). Upon application of McFadden's [18] more rigorous fold test, however, the test is failed at the 95% confidence level (unfolded SCOS1 = 7.525 > 4.358, the 95% critical value) when the sampled three fold limbs are considered, despite precision parameter, κ, reaching maximum at 100% unfolding. Applying the same test

Table 1
Silurian site statistics for component C at Shiqian, Guizhou (27.5°N, 108.0°E)

Site	Bedding	n/N	In situ		Tilt-corrected				Pole position	
			D (°)	I (°)	D (°)	I (°)	α_{95}	κ_s	Lat. (°N)	Long. (°E)
6[a]	174/45	5/5	77.7	−47.4	79.9	−2.6	9.0	154.0	8.3	203.9
7	174/45	7/7	67.6	−21.3	67.5	22.0	7.5	65.6	25.1	197.8
8	34/39	6/6	43.4	52.3	77.7	33.5	4.5	222.4	18.9	186.7
9	34/39	7/7	63.6	39.7	80.2	14.9	5.4	125.4	12.2	195.7
10	34/39	8/8	61.3	43.7	81.1	19.2	3.3	277.0	12.4	193.2
11	34/39	8/8	57.6	36.1	74.0	14.7	3.4	269.8	17.6	198.7
12	34/39	3/3	54.3	39.1	73.8	18.7	5.6	478.4	18.7	196.7
13	34/39	6/6	54.5	41.0	75.2	20.1	7.2	88.5	17.8	195.4
14	34/39	6/6	78.4	34.8	87.9	4.7	7.9	73.5	2.9	196.9
15[a]	223/54	7/7	54.5	−18.3	64.5	−1.8	5.6	170.8	22.0	211.3
16[a]	223/54	5/5	57.6	−17.2	65.4	1.2	15.9	50.3	22.0	209.4
17[a]	223/54	7/7	59.6	−19.1	68.0	1.4	5.9	153.8	19.7	207.9
18[a]	221/51	5/7	51.7	−16.7	60.6	−2.4	13.3	71.8	25.2	213.7
19[a]	221/51	6/7	65.3	−27.4	76.8	−0.3	14.2	38.1	11.6	204.3
Global mean		14/19	60.5	8.9	73.7	10.3			16.9	200.7
			$\kappa_g = 5.4$		$\kappa_s = 34.6$		$\alpha_{95} = 6.9$		$A_{95} = 4.8$	
Mean of 6–14 (C1)[b]		9/	62.5	27.2	77.6	16.2			14.9	196.1
			$\kappa_g = 5.7$		$\kappa_s = 48.1$		$\alpha_{95} = 7.5$		$A_{95} = 5.1$	
Mean of 8–19		12/	58.7	16.3	73.6	10.4			16.9	200.7
			$\kappa_g = 6.4$		$\kappa_s = 34.5$		$\alpha_{95} = 7.5$		$A_{95} = 5.4$	

Bedding: strike/dip; n/N: numbers of samples (sites) used/measured; D, I: declination and inclination; α_{95}, A_{95}: radius of the 95% confidence circle about the mean direction and pole; κ_g, κ_s: Fisher precision parameters before and after tilt correction.
[a]Site mean direction determined using the sector-constrained great circle intersection method [13].
[b]See text for details.

to the two limbs of the syncline (sites 6–14), the test is positive (in situ SCOS1 = 7.139, unfolded SCOS1 = 1.838, 95% critical value = 3.497); to the two limbs of the anticline (sites 8–19), the test is negative (in situ SCOS1 = 9.018, unfolded SCOS1 = 7.499, 95% critical value = 4.036). The tilt-corrected mean directions of sites 15–19 from the western limb of the anticline appear to be shallower and less easterly than those of sites 6–14 from the other two limbs. F-test [19] indicates that these two groups of directions are indeed statistically distinguishable (F-statistic = 11.67 > 3.40 = F(2, 24), $R_1 = 8.834$, $R_2 = 4.976$, $R = 13.624$, $n = 14$). Interestingly, the tilt-corrected mean direction of sites 15–19 from Shiqian ($D/I = 67.1°/-0.4°$, $\alpha_{95} = 6.0°$) is nearly identical to that of the C component from Daguan ($D/I = 64.6°/-0.3°$, $\alpha_{95} = 4.5°$); in fact, if these sites are considered as a group, they pass McFadden's [18] fold test at the 95% confidence level (in situ SCOS1 = 12.141, unfolded SCOS1 = 0.318, 95% critical value = 4.358, $n = 14$). Thus, two groups of the C component directions appear to have been recovered: one (C1 hereafter) includes sites 6–14 from Shiqian which yields an overall tilt-corrected mean direction of $D/I = 77.6°/16.2°$ ($\alpha_{95} = 7.5°$) and a pole at 14.9°N/196.1°E ($A_{95} = 5.1°$) (see Table 1), another (C2 hereafter) includes sites 15–19 from Shiqian and nine sites from Daguan which yields a mean of $D/I = 65.4°/-0.3°$ ($\alpha_{95} = 3.3°$) and a pole at 21.5°N/207.5°E ($A_{95} = 2.9°$) (see Table 2). Both groups of directions are pre-folding and were probably acquired before late Cretaceous time.

In most of those sites, the mean directions of which were determined from great circle intersections, the great circles are sub-parallel. To check if the distinction between the C1 and C2 directions is due to the utilization of such a technique, we tried the method suggested by Enkin et al. [20]. We first computed an average great circle and associated sector constraints for each of these sites

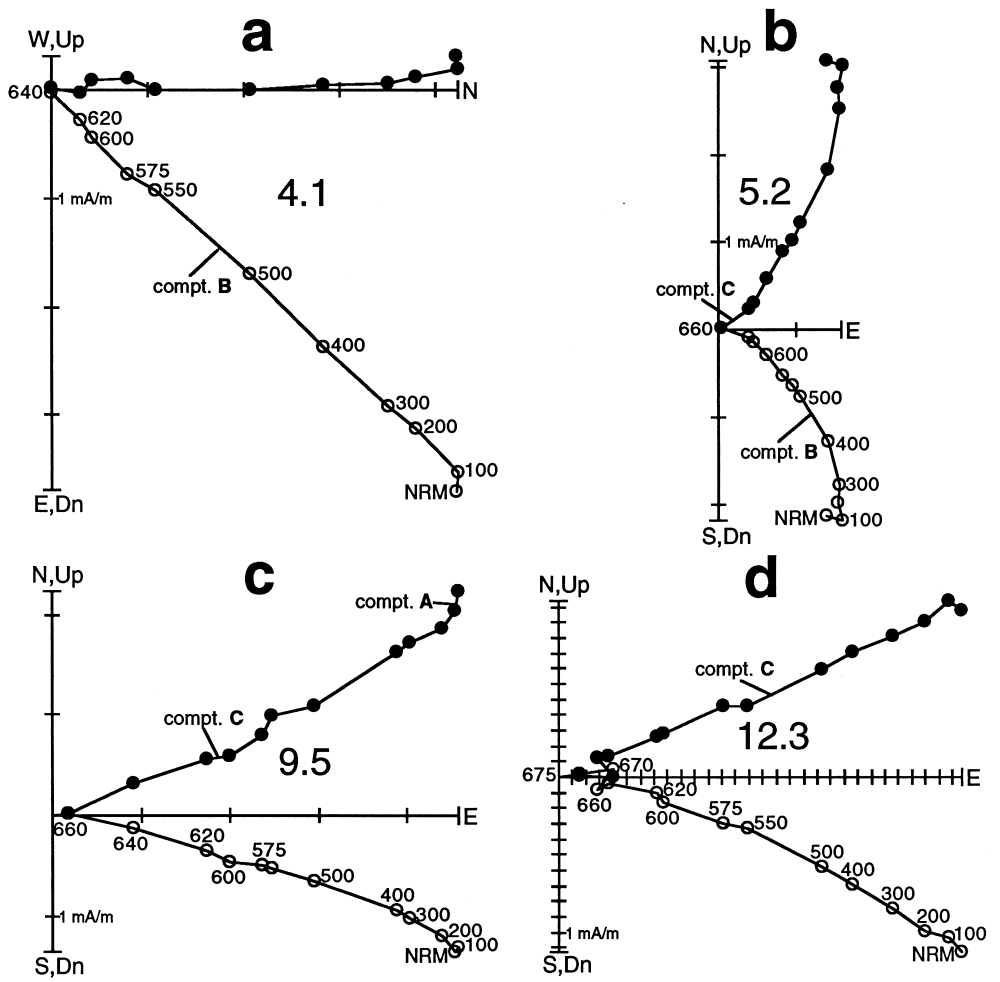


Fig. 5. Representative orthogonal plots of thermal demagnetization for the samples from Daguan. Conventions as in Fig. 2.

using Fisherian statistics [10], and then applied McFadden and McElhinny's combined analysis of remagnetization circles and direct observations [13] to the means of all usable sites. As can be seen from Fig. 7, the mean great circles of sites 15–19 from Shiqian barely pass the site mean directions determined using principal component analysis technique [12]; by contrast, these great circles pass right through the mean directions of the sites from Daquan, yielding a much tighter distribution after tilt correction. Thus, we believe that the C1 and C2 are indeed distinct groups of directions. The mean tilt-corrected C1 and C2 directions computed this way are $D/I = 76.5°/16.6°$ ($\alpha_{95} = 7.5°$, $\kappa = 48.6$) and $D/I = 66.0°/-1.0°$ ($\alpha_{95} = 2.9°$, $\kappa = 203.1$), respectively, which are nearly identical to those listed in Tables 1 and 2. We will use the latter for further discussion.

What might then be responsible for the difference between these two groups of the C component directions? One possibility is rotation about vertical axis between the sampling sites. This seems unlikely, though, given that the difference does not only involve declination but also inclination and that the C2 direction is present at both sampling localities. Another possibility is that the two groups of directions represent magnetizations acquired at different times. This possibility ap-

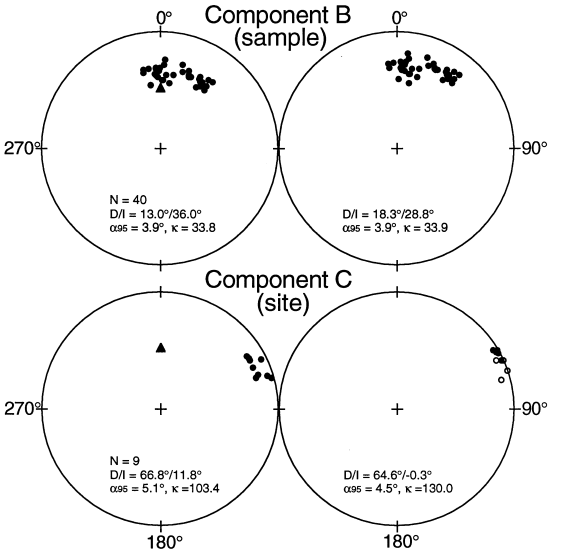


Fig. 6. Equal-area projections of components B and C resolved from the samples from Daguan. Conventions as in Fig. 3.

pears to gain support from the apparent polar wander path (APWP) for the YB (Fig. 8). The 95% confidence circle (A_{95}) about the C1 paleomagnetic pole overlaps that of the Silurian pole derived from Xiushan, southern Sichuan, about 150 km northeast of Shiqian, which also passes a fold test [1]. Such concordance gives us some confidence that the C1 magnetization observed at these two localities was acquired at about the same time, probably in the early history of the rocks. The C2 paleomagnetic pole lies on the track of the APWP between the C1 pole and the Carboniferous and Permian poles, suggesting that the C2 magnetization might be acquired sometime earlier than the Carboniferous but later than the C1 magnetization.

4.2. Paleogeographic implications

A simplified APWP for the YB is shown in Fig. 8. Since our attention is focused on the Paleozoic era, only the better-determined late Cretaceous and late Jurassic poles [21] are indicated for late Mesozoic times. A recalculated mean early Triassic pole is used, for some of the reported early Triassic poles may have been affected by local rotations [22]. Reported late Permian poles are abundant, but significant differences exist between them. As we argued elsewhere, local rotations may be the main cause for such differences [23]. The difficulty is that we do not know yet which poles were affected by rotations. It seems that the segment of the APWP from Permian to Triassic is smoother if the poles computed from the north-

Table 2
Silurian site statistics for component C at Daguan, Yunnan (27.9°N, 103.9°E)

Site	Bedding	*n*/*N*	In situ		Tilt-corrected				Pole position	
			D (°)	*I* (°)	*D* (°)	*I* (°)	α_{95}	κ_s	Lat. (°N)	Long. (°E)
5[a]	335/11	6/6	60.4	14.7	60.5	3.7	10.6	68.4	26.7	207.0
6[a]	335/11	5/7	58.6	15.9	58.8	5.0	15.1	55.7	28.5	207.3
8	335/11	6/6	74.5	2.5	74.6	−8.4	5.5	152.0	11.5	205.0
9	330/13	5/6	64.0	6.0	64.0	−7.0	9.3	68.1	21.0	210.0
10	330/13	6/6	61.3	15.1	61.2	2.1	7.4	83.5	25.7	207.3
11	267/43	6/6	72.3	16.3	65.3	2.2	9.7	48.8	22.2	205.0
12	267/43	5/5	70.8	13.7	65.9	−0.7	5.2	219.3	21.0	206.0
13	261/42	6/6	72.8	5.6	71.2	−1.4	7.3	86.2	16.2	203.6
14	261/42	6/7	66.0	15.6	59.6	1.9	4.4	230.4	27.1	208.3
Global mean		9/14	66.8	11.8	64.6	−0.3			22.2	206.6
			$\kappa_g = 103.4$		$\kappa_s = 130.0$		$\alpha_{95} = 4.5$		$A_{95} = 3.7$	
C2 mean[b]		14/	63.6	0.6	65.4	−0.3			21.5	207.5
			$\kappa_g = 20.9$		$\kappa_s = 146.6$		$\alpha_{95} = 3.3$		$A_{95} = 2.9$	

Explanations as in Table 1.
[a]Site mean direction determined using the sector-constrained great circle intersection method [13].
[b]Mean direction computed with the inclusion of sites 15–19 from Shiqian, see text for details.

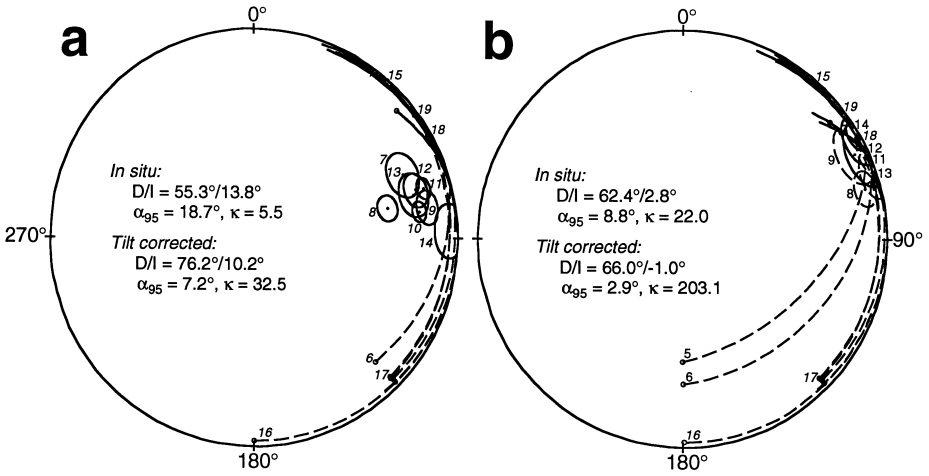


Fig. 7. Combined analysis of mean constrained great circles and vector end point directions for the sites used (in stratigraphic coordinates). (a) Sites from Shiqian. Note that the mean great circles barely pass the mean site directions determined using principal component analysis. (b) Sites from Daguan plus sites 15–19 from Shiqian. The tilt-corrected mean great circles and mean site directions yield a tight distribution. Solid (dashed) lines are projections on lower (upper) hemisphere. Sites from Shiqian are indicated with numbers in italics next to the confidence circles or mean constrained great circles.

eastern declinational directions are used. Therefore, we choose to use the pole we recently derived from an Emeishan basalt section [23] instead of using a mean pole. The late Carboniferous pole reported by Lin et al. [2] is preliminary with fewer samples and larger uncertainties, but it appears to be on the track between the Silurian poles and the late Permian pole. Bai et al. [24] recently obtained a pole from the mid-Cambrian redbeds from northern Sichuan, the reliability of which is supported by a larger collection, thorough demagnetization and a positive fold test. The paleomagnetic pole reported by Fang et al. [3] from the Devonian rocks from eastern Yunnan also seems to lie along the track from the mid-Cambrian pole to the late Carboniferous pole. If this pole represents a genuine Devonian pole for the YB, the apparent polar wander would be as much as 45° from late Devonian to late Carboniferous and the C1 and C2 magnetizations observed from the Silurian formations from three widely separated localities would probably be post-Devonian remagnetizations. However, the Devonian results are based on a smaller number of samples with no field test. Therefore, further work is needed to verify the Devonian results.

At present time, we consider the C1 magnetiza-

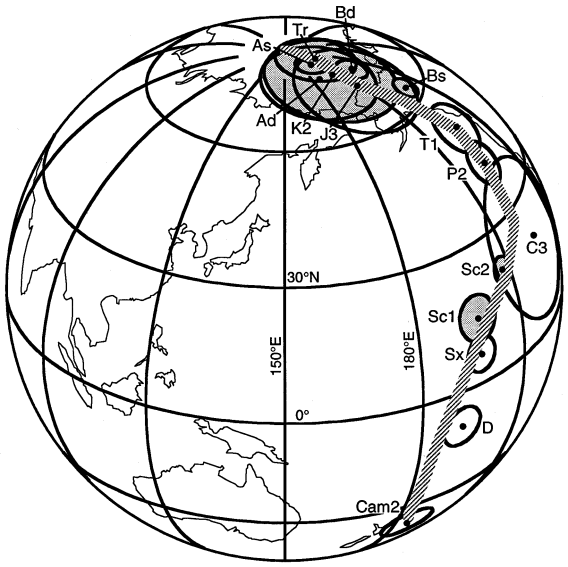


Fig. 8. APWP (hatched) for the YB. Paleomagnetic poles used: Cam2 = mid-Cambrian [24], Sx = Silurian from Xiushan, southern Sichuan [1], C3 = late Carboniferous [2], P2 = late Permian [23], T1 = early Triassic [22], J3 = late Jurassic [21], K2 = late Cretaceous [21], Tr = Tertiary [28]. Poles computed from different magnetic components of this study are shaded: Sc1 = C1, Sc2 = C2, Bs = the B component from Shiqian, Bd = the B component from Daguan, As = the A component from Shiqian, Ad = the A component from Daguan. Also indicated is the pole (D) reported by Fang et al. [3] from the Devonian rocks.

tion to be best interpreted as a primary remanence acquired during the Silurian and the C2 magnetization acquired sometime between the Carboniferous and Silurian. Thus, the mean inclinations translate into a paleolatitude of $8.3 \pm 4.0°$ during the Silurian and $0.2 \pm 2.0°$ probably during the Devonian for the sampling localities. Furthermore, the APWP for the YB indicate that the paleopoles derived from these magnetizations are likely to be north poles (Fig. 8); therefore, northeastern Guizhou (Shiqian) was in the northern equatorial area during the Silurian. This reinforces the view held by most paleomagnetists that the YB had been equatorial probably for the entire Paleozoic. During this period, the YB may have wandered around the paleo-equator, for some period of time being to the north of the equator such as in the Silurian and for some other period of time being to the south of the equator such as in the Middle Cambrian and late Permian. Some authors suspect that the shallow inclinations commonly observed in the Paleozoic rocks from the YB are a sign of remagnetization [4]. Although legitimate, such suspicion is probably premature. First, the paleomagnetically determined equatorial or low latitudes for the YB agree well with the paleoclimatic data [25]. Second, although the paleolatitudes observed for the YB are consistently low for almost the entire Paleozoic, the YB appear to have been rotating in a clockwise fashion since the Middle Cambrian (Fig. 8). Using the pole data in Fig. 8, a clockwise rotation of $87.7° \pm 9.6°$ for the YB is estimated for the period from mid-Cambrian through late Permian. This trend appears to have continued into early Mesozoic and is certainly compatible with the scissors-like collision model between the YB and the North China Block (NCB) proposed by Zhao and Coe [26], in which the collision first took place in the east and then progressed westward through clockwise rotation of the YB relative to the NCB.

5. Conclusion

Progressive thermal demagnetization revealed multiple magnetic components from the Silurian redbed samples: a low unblocking temperature component (A) of recent origin, a middle to high unblocking temperature component (B) probably acquired during late Mesozoic time, and a pre-folding, high unblocking temperature component (C). Two statistically distinct groups of paleomagnetic directions were further recognized from the C component. Both groups of C directions are of shallow inclinations but one (C1) has more easterly declinations than the other (C2). The APWP for the YB suggests that acquisition of the C1 magnetization was earlier than that of the C2 magnetization; the former was probably acquired during the Silurian and the latter was acquired sometime between the Silurian and Carboniferous. The shallow inclinations of the C component reconfirm the equatorial position for the YB during the mid-Paleozoic, and the declinations indicate a continuous clockwise rotation of the YB since at least the mid-Cambrian. *[CL]*

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